momently

Quality Requirements

CYBERNAUTS

Non-Functional Requirements & Architectural Design Decisions for Momently

This document specifies the non-functional and quality requirements for the Momently system. Each requirement is uniquely identified, assigned to an architectural quality attribute category, stated as a verifiable condition, and accompanied by an explicit, measurable target metric. Requirements are prioritized as **High, Medium, or Low** to align with execution phases and enterprise service level agreements (SLAs).

1. Quality Attribute Matrix

1.1 Performance Requirements

ID	Requirement / Priority	Verifiable Condition	Acceptance Metric Target
QR-PE-01	**Response Time** [HIGH]	The system shall respond to all standard user interactions (including page loads, time entry submissions, and timesheet views) within 2 seconds under normal operating conditions.	95th percentile response time ≤ 2s under a load of 100 concurrent users.
QR-PE-02	**Response Time (Reports)** [HIGH]	The system shall respond to report generation requests within 5 seconds for datasets spanning up to 12 months.	Report generation ≤ 5s for up to 1 year of data per workspace.
QR-PE-03	**Throughput** [HIGH]	The system shall support a minimum of 500 simultaneous authenticated users performing time entry and timesheet operations without performance degradation.	500 concurrent users; no degradation beyond 10% of baseline response time.

QR-PE-04	**Caching** [MEDIUM]	Frequently accessed data (API responses, user session data, summaries) shall be served from the Redis cache to minimize backend database load.	Cache hit rate ≥ 70% for eligible endpoints; DB query reduction ≥ 40%.
QR-PE-05	**AI Insight Latency** [MEDIUM]	AI-generated insights (burnout risk, anomaly detection, productivity scores) shall be computed asynchronously. Processing is deferred to Demo 2.	AI inference latency ≤ 10s end-to-end for standard prediction workloads.

1.1 Reliability Requirements

ID	Requirement / Priority	Verifiable Condition	Acceptance Metric Target
QR-RE-01	**Availability** [HIGH]	The system shall maintain 99% uptime during defined business hours (08:00-18:00, Monday-Friday). Scheduled maintenance windows shall be communicated explicitly in advance.	Uptime ≥ 99% (measured monthly); max unplanned downtime ≤ 7.2 hrs/month.
QR-RE-02	**Fault Tolerance** [HIGH]	The system shall gracefully handle the failure or latency of non-critical external third-party components (Jira, GitHub, Calendar APIs). Core time-tracking functionality shall remain uninterrupted.	Core time entry and approval flows remain operational during external integration outages.
QR-RE-03	**Disaster Recovery** [HIGH]	In the event of a critical infrastructure or database failure, the	Recovery Time Objective (RTO) ≤ 2 hours; Recovery

		system shall be fully restorable within two hours, ensuring minimal data loss.	Point Objective (RPO) = 0 (no committed transaction loss).
QR-RE-04	**Data Integrity** [HIGH]	All time entry, timesheet status changes, and audit log entries shall be persisted with full transactional integrity. Partial writes must be systematically rejected.	Zero partial-write incidents; all DB operations wrapped in strict SQL ACID transactions.
QR-RE-05	**Error Handling** [MEDIUM]	The system shall surface clear, user-friendly error messages for all foreseeable failure states. Highly structured error details shall be logged server-side for internal diagnostics.	All user-facing errors include a clear message, error code, and suggested action; errors logged with timestamp, user ID, and stack trace.

1.1 Scalability Requirements

ID	Requirement / Priority	Verifiable Condition	Acceptance Metric Target
QR-SC-01	**Horizontal Scaling** [HIGH]	The system architecture shall inherently support horizontal scaling of the Spring Boot backend and Python AI microservices via container orchestration platforms.	New service instances deployable via Docker/AWS ECS with zero application code changes required.
QR-SC-02	**Data Volume Resilience** [HIGH]	The system shall fully maintain core performance benchmarks (QR-PE-01, QR-PE-02) as the primary time entries table grows significantly over time.	No performance regression observed at 1,000,000 time entry records compared to a 10K baseline sample.

QR-SC-03	**User Growth** [MEDIUM]	The workspace infrastructure shall seamlessly accommodate corporate user growth from the initial test team size up to 1,000 registered users without foundational refactoring.	System handles 1,000 users with ≤ 15% increase in baseline 95th-percentile response time.
QR-SC-04	**AI Workload Isolation** [MEDIUM]	AI/ML microservices shall run and be scaled completely independently from the main operational backend application.	AI services deployable and scalable independently; scaling AI workloads introduces zero degradation to core API SLAs.

1.1 Security Requirements

ID	Requirement / Priority	Verifiable Condition	Acceptance Metric Target
QR-SE-01	**Authentication & MFA** [HIGH]	The system shall strictly enforce two-factor authentication via one-time passwords (OTP) for all user logins. Credentials must map to the organization's company domain.	OTP required for all logins; OTP tokens automatically expire after 5 minutes; non-company domains rejected.
QR-SE-02	**Authorization (RBAC)** [HIGH]	The system shall strictly enforce granular role-based access control (RBAC) across every API endpoint. Cross-workspace data isolation must be guaranteed.	100% of REST API endpoints protected; unauthorized requests immediately rejected with an HTTP 403 Forbidden.

QR-SE-03	**Session Management** [HIGH]	User authorization sessions shall be securely managed via stateless JSON Web Tokens (JWT).	Token Time-To-Live (TTL) = 1 hour of inactivity; refresh tokens instantly invalidated upon user logout; zero session state stored server-side.
QR-SE-04	**Data in Transit** [HIGH]	All client-to-server and inter-service communications shall be fully encrypted using Transport Layer Security.	TLS 1.2 or higher enforced globally across all network endpoints; plain unencrypted HTTP connections blocked or redirected.
QR-SE-05	**Data at Rest (Secrets)** [HIGH]	Sensitive system configurations, third-party API tokens, credentials, and database passwords shall never be hardcoded into source repositories.	Zero hardcoded secrets allowed in codebase; 100% of runtime secrets retrieved securely from AWS Parameter Store.
QR-SE-06	**Immutable Audit Logging** [HIGH]	The system shall maintain an unalterable, structured audit log recording all major transactional and administrative modifications.	Audit records written for 100% of defined events (edits, approvals, roles); logs are strictly append-only and tamper-evident.
QR-SE-07	**POPIA Compliance** [HIGH]	The system shall store, handle, and process personal information strictly in accordance with the South African Protection of Personal Information Act (POPIA).	Data access and export operations require explicit authorization; zero PII data exported without a secure, immutable audit trail.

1.1 Maintainability Requirements

ID	Requirement / Priority	Verifiable Condition	Acceptance Metric Target
QR-MA-01	**Code Quality Gates** [HIGH]	All source code shall adhere strictly to uniform team coding standards. Automated static analysis tools must actively validate rules within the CI/CD pipeline.	Zero critical static analysis or quality gate violations allowed on any merge to the main production branch.
QR-MA-02	**Automated Test Coverage** [HIGH]	The backend application must preserve an extensive test suite ensuring major features do not suffer regressions.	Backend unit test line coverage ≥ 80%; 100% of public REST endpoints covered comprehensively by integration tests.
QR-MA-03	**System Documentation** [MEDIUM]	All external and internal REST API endpoints shall be comprehensively mapped. Architectural overviews and data models must be fully maintained.	100% of functional endpoints documented live in Swagger/OpenAPI; core documentation maintained inside markdown files in the repository.
QR-MA-04	**Modifiability & Design** [HIGH]	The codebase shall utilize a distinct layered or hexagonal architecture separating core responsibilities (presentation, domain logic, data access, AI inference).	Changes to isolated layers demand zero overlapping structural modifications in adjacent layers during code reviews.
QR-MA-05	**CI/CD Deployment Pipeline** [HIGH]	A thoroughly automated CI/CD pipeline must govern the software compilation, verification, packaging, and staging deployment cycles.	Pipeline executes natively on every push; production deployments structurally blocked without verified test suites and peer code reviews.

QR-MA-06	**Environment Configuration** [MEDIUM]	The compiled application artifacts shall natively support deployment across development, staging, and production environments without rebuilding code.	Deployable across any environmental tier via dynamic configurations alone; zero environment-specific logic in the source code.

1.1 Usability Requirements

ID	Requirement / Priority	Verifiable Condition	Acceptance Metric Target
QR-US-01	**User Learnability** [MEDIUM]	A newly onboarded user with zero prior product training shall be able to easily create a time entry, submit a completed timesheet, and extract a basic report.	User experience testing: ≥ 80% of test participants successfully execute all three core tasks within 10 minutes entirely unaided.
QR-US-02	**Accessibility Standards** [MEDIUM]	The web-based application interface shall comply clearly with international digital accessibility guidelines.	WCAG 2.1 Level AA compliance verified regularly using automated industry-standard accessibility audit tooling.
QR-US-03	**Responsive UI Layouts** [MEDIUM]	The application UI viewport shall remain perfectly fluid, highly functional, and visually balanced across standard hardware display profiles.	Zero layout breakages, overlapping text, or interactive function loss when tested at 1920x1080 (Desktop), 1366x768 (Laptop), and 768px wide (Tablets).
QR-US-04	**Theme Personalization** [LOW]	The user interface shall allow individual users to switch seamlessly	Visual theme selection persisted securely per user profile; applied

		between stylized light and dark display modes.	instantly across sessions upon initial authentication.

1.1 Interoperability Requirements

ID	Requirement / Priority	Verifiable Condition	Acceptance Metric Target
QR-IN-01	**External API Integrations** [HIGH]	The platform shall seamlessly integrate with enterprise external services (Jira, GitHub, and Google/Microsoft Calendar APIs) utilizing non-blocking clients.	External API network failures handled via standardized resilient circuit-breaker patterns; core system features remain completely operational.
QR-IN-02	**Enterprise SSO Compatibility** [HIGH]	The security layer shall be designed from day one to quickly support identity management protocols for enterprise Single Sign-On.	SSO capability integration achievable purely via configuration property updates; zero backend architectural refactoring or code alterations required.
QR-IN-03	**REST API Architecture Standards** [MEDIUM]	All public and internal microservice APIs shall conform strictly to uniform architectural principles and maintain explicit versioning rules.	API design validated against REST Maturity Model Level 2+ parameters; strict version control enforced natively via explicit URL path prefixes (e.g., /api/v1/).

2. Architectural & Key Design Decisions

Performance Optimization: Redis Caching & Asynchronous AI Workloads

- The strict 2-second standard response target (QR-PE-01) stems directly from corporate user needs for a highly efficient internal tool tracking system to replace the legacy TMetric platform.
- High-frequency operational reads, frequent external platform synchronized data blocks (e.g., Jira issues, GitHub commits), and active user sessions will be captured cleanly by a distributed Redis caching layer (QR-PE-04).
- Heavier analytical operations, such as predictive burnout mapping, work-pattern anomalies, and automated performance profiling (QR-PE-05), are structurally decoupled to evaluate completely asynchronously, ensuring that background model execution times never introduce latency onto active API data loops. These deep analytical services are planned for final delivery in Demo 2.

Reliability Framework: ACID Database Guarantees and Resilient RTO/RPO

- Operating directly within an institutional financial services landscape dictates that all active data streams and auditing records preserve perfect synchronization across historical timelines.
- The core application utilizes advanced PostgreSQL ACID configurations (QR-RE-04) combined with an optimized 2-hour target Disaster Recovery framework (QR-RE-03) to satisfy corporate operational risk boundaries.
- To isolate downstream network volatility, the integration architecture wraps all connections to third-party endpoints (such as GitHub commits or Calendar feeds) in automated software circuit-breaker design patterns (QR-RE-02), maintaining uninterrupted primary time-logging capabilities even if external vendor web services encounter critical service degradation.

Security & Regulatory Strategy: POPIA Compliance, MFA, and Stateless Authorization

- As a modern corporate entity handling financial operations, the client must conform tightly to the rules established by South African Protection of Personal Information Act (POPIA) mandates.
- Requirement QR-SE-07 sets formal operational standards governing customer data access boundaries, restricted export capabilities, and unified tracking logs for personal data modifications.
- Primary edge security implements strict Multi-Factor Authentication handled via email-based One-Time Passwords (OTP) supported by a secure JavaMailSender

integration using Mailtrap infrastructure, enforcing automatic 5-minute code validation timeouts (QR-SE-01).

- Subsequent REST requests are validated independently via stateless JSON Web Tokens (JWT) coupled directly with Spring Security Role-Based Access Controls (RBAC), satisfying strict zero-trust operational requirements while maintaining high execution efficiency. The authentication architecture natively enables straightforward migration to centralized corporate identity providers via OAuth2 / OpenID Connect profiles (QR-IN-02) down the road without structural rework.

Maintainability Architecture: Structural Hexagonal Layers and CI/CD Automation

- The structural layout relies on a clear layered software design pattern separating specialized domains cleanly (Angular UI presentation, Spring Boot backend components, and dedicated Python AI inference microservices) so they can scale and evolve independently.
- A baseline automated software line test coverage standard of 80% (QR-MA-02) integrated into an automated GitHub Actions deployment pipeline (QR-MA-05) serves as the core quality gate for active Agile Scrum cycles.
- Every application pull request must achieve complete compilation, validate architectural check constraints, and secure formal peer code review validation before automated deployments are initiated, ensuring a consistent, production-ready artifact throughout the development lifecycle.

Elastic Scalability: Isolated Artificial Intelligence Workloads

- By decoupling compute-intensive AI operations out into separate, standalone Python microservices (QR-SC-04), the framework isolates heavier processor-bound routines away from core operational endpoints.
- This guarantees that spikes in analytics traffic never impact routine daily interactions such as clocking hours or generating weekly team sheets.